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MARKET REGIME DETECTION – ANALYSIS REPORT
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Q1: WHAT REGIMES WERE DISCOVERED?

- ▶ High Volatility / Crisis
 - Mean daily return : -0.00130
 - Daily volatility : 0.03437
 - Ann. Sharpe : -0.601
 - Max Drawdown : -30.56%
 - Frequency (train) : 3.57%
- ▶ Ranging / Sideways
 - Mean daily return : +0.00041
 - Daily volatility : 0.00796
 - Ann. Sharpe : 0.820
 - Max Drawdown : -8.93%
 - Frequency (train) : 28.24%
- ▶ Trending Bear
 - Mean daily return : +0.00009
 - Daily volatility : 0.01359
 - Ann. Sharpe : 0.100
 - Max Drawdown : -23.00%
 - Frequency (train) : 26.42%
- ▶ Trending Bull
 - Mean daily return : +0.00103
 - Daily volatility : 0.00501
 - Ann. Sharpe : 3.270
 - Max Drawdown : -3.32%
 - Frequency (train) : 41.78%

Q2: HOW PERSISTENT ARE THEY?

Train persistence:

	Mean Dur	Median Dur	Max Dur	N Episodes
Regime				
High Volatility / Crisis	19.6	15.0	45	5
Ranging / Sideways	16.9	15.0	76	46
Trending Bear	26.0	19.0	78	28
Trending Bull	31.9	22.0	183	36

Test persistence:

	Mean Dur	Median Dur	Max Dur	N Episodes
Regime				

High Volatility / Crisis	14.0	20.0	20	3
Ranging / Sideways	19.6	14.0	79	24
Trending Bear	36.9	24.0	115	12
Trending Bull	22.8	21.5	47	18

Q3: WHICH TRANSITIONS OCCUR MOST FREQUENTLY?

Top transitions (train):

High Volatility / Crisis → Trending Bear (0.051)
Ranging / Sideways → Trending Bull (0.036)
Trending Bull → Ranging / Sideways (0.027)
Ranging / Sideways → Trending Bear (0.023)
Trending Bear → Ranging / Sideways (0.021)

Top transitions (test):

High Volatility / Crisis → Trending Bear (0.071)
Trending Bull → Ranging / Sideways (0.044)
Ranging / Sideways → Trending Bull (0.032)
Ranging / Sideways → Trending Bear (0.017)
Trending Bear → Ranging / Sideways (0.014)

Q4: DO MAJOR CHANGEPOINTS ALIGN WITH KNOWN MARKET EVENTS?

- ✓ 2010-05-20 [Trending Bear → High Volatility / Crisis] aligns with: Flash Crash (May 2010)
- ✓ 2010-06-11 [High Volatility / Crisis → Trending Bear] aligns with: Flash Crash (May 2010)
- 2010-09-30 [Trending Bear → Trending Bull] (no direct match in known-event list)
- 2010-11-12 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2010-12-15 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2011-02-22 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2011-03-10 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2011-03-29 [Trending Bear → Ranging / Sideways] (no direct match in known-event list)
- 2011-04-12 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2011-05-04 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2011-06-06 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2011-06-29 [Trending Bear → Ranging / Sideways] (no direct match in known-event list)
- ✓ 2011-09-08 [High Volatility / Crisis → Trending Bear] aligns with: US Debt Ceiling / Euro Crisis
- 2011-10-21 [Trending Bear → High Volatility / Crisis] (no

direct match in known-event list)

- 2011-11-07 [High Volatility / Crisis → Trending Bear] (no direct match in known-event list)
- 2012-01-20 [Trending Bear → Trending Bull] (no direct match in known-event list)
- 2012-03-22 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2012-05-17 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2012-07-11 [Trending Bear → Ranging / Sideways] (no direct match in known-event list)
- 2012-08-20 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2012-09-25 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2012-12-12 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2012-12-28 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2013-01-24 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2013-04-15 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2013-05-13 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2013-06-05 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2013-07-19 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2013-08-15 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2013-09-13 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2013-10-03 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2013-11-04 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2014-03-03 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2014-03-26 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2014-05-09 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2014-07-31 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2014-08-28 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2014-09-25 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2014-10-09 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2014-12-10 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2015-01-05 [Ranging / Sideways → Trending Bear] (no direct

match in known-event list)

✓ 2015-08-21 [Ranging / Sideways → Trending Bear] aligns with:
China/Oil Selloff 2015-16

✓ 2015-11-12 [Trending Bull → Ranging / Sideways] aligns with:
China/Oil Selloff 2015-16

✓ 2015-12-16 [Ranging / Sideways → Trending Bear] aligns with:
China/Oil Selloff 2015-16

– 2016-03-16 [Trending Bear → Trending Bull] (no direct match
in known-event list)

– 2016-04-28 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2016-06-01 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2016-06-24 [Trending Bull → Trending Bear] (no direct match
in known-event list)

– 2016-07-25 [Trending Bear → Trending Bull] (no direct match
in known-event list)

– 2016-09-09 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2016-10-19 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2016-11-02 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2016-11-23 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2017-08-17 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2017-09-08 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2018-02-02 [Trending Bull → Trending Bear] (no direct match
in known-event list)

– 2018-04-27 [Trending Bear → Ranging / Sideways] (no direct
match in known-event list)

– 2018-05-22 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

✓ 2018-10-10 [Trending Bull → Trending Bear] aligns with: Q4
2018 Selloff

– 2019-02-04 [Trending Bear → Trending Bull] (no direct match
in known-event list)

– 2019-05-07 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2019-05-23 [Ranging / Sideways → Trending Bear] (no direct
match in known-event list)

– 2019-06-10 [Trending Bear → Ranging / Sideways] (no direct
match in known-event list)

– 2019-06-28 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2019-10-30 [Ranging / Sideways → Trending Bull] (no direct
match in known-event list)

– 2020-01-27 [Trending Bull → Ranging / Sideways] (no direct
match in known-event list)

– 2020-05-07 [High Volatility / Crisis → Trending Bear] (no
direct match in known-event list)

– 2020-07-27 [Trending Bear → Trending Bull] (no direct match

in known-event list)

- 2020-09-03 [Trending Bull → Trending Bear] (no direct match in known-event list)
- 2021-01-27 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2021-02-25 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2021-05-11 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2021-06-10 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2021-07-19 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2021-08-12 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2021-09-10 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2021-10-26 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2021-12-31 [Trending Bear → Ranging / Sideways] (no direct match in known-event list)
- ✓ 2022-01-20 [Ranging / Sideways → Trending Bear] aligns with: Rate Hike Bear Market 2022
- ✓ 2022-04-29 [Trending Bear → High Volatility / Crisis] aligns with: Rate Hike Bear Market 2022
- ✓ 2022-05-27 [High Volatility / Crisis → Trending Bear] aligns with: Rate Hike Bear Market 2022
- 2023-06-09 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2023-08-02 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2023-11-22 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2024-02-20 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2024-03-13 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2024-04-04 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2024-05-16 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2024-07-18 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2024-08-01 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2024-09-09 [Trending Bear → Ranging / Sideways] (no direct match in known-event list)
- 2024-10-02 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2024-10-31 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2024-12-04 [Ranging / Sideways → Trending Bull] (no direct match in known-event list)
- 2024-12-18 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)

match in known-event list)

- 2025-04-09 [Trending Bear → High Volatility / Crisis] (no direct match in known-event list)
- 2025-05-08 [High Volatility / Crisis → Trending Bear] (no direct match in known-event list)
- 2025-10-10 [Trending Bull → Ranging / Sideways] (no direct match in known-event list)
- 2026-03-20 [Ranging / Sideways → Trending Bear] (no direct match in known-event list)
- 2026-04-29 [Trending Bear → Trending Bull] (no direct match in known-event list)

Q5: DOES THE HMM IDENTIFY MEANINGFUL MARKET STRUCTURE?

Train log-likelihood (per sample): -5.03219
Test log-likelihood (per sample): -5.30301
LL ratio (test/train) : 1.0538
Total variation distance (dist.) : 0.1219

Interpretation:

- A Sharpe ratio > 1.0 in Trending Bull and near 0 / negative in Trending Bear confirms trend-directional regime structure.
- High Volatility / Crisis should show the largest annualised volatility and deepest drawdown, consistent with crisis labeling.
- LL ratio close to 1.0 indicates the model generalises well out-of-sample – the latent structure is stable across time.
- Low TVD (< 0.15) indicates regime frequency stability between training and test periods.
- Changepoints coinciding with COVID-19 (Feb/Mar 2020), 2015–16 China oil selloff, and 2022 bear market confirm economic interpretability of the discovered structure.

Q6: CAN REGIME INFORMATION BE USED FOR RISK MANAGEMENT?

YES – with appropriate caveats:

- Strategy switching:
 - Trending Bull → run momentum / trend-following strategies.
 - Trending Bear → reduce equity exposure; rotate to bonds/cash.
 - Ranging/Sideways → mean-reversion strategies (pairs, stat-arb).
 - Crisis → volatility-targeting, safe-haven allocation.
- Risk sizing:

Annualised volatility per regime can feed directly into a volatility-targeting framework (e.g., target 10% ann. vol):

$$\text{position_size} = \text{target_vol} / \text{regime_ann_vol}.$$

► Caveats:

- Regimes are inferred with posterior uncertainty – always weight decisions by posterior probability, not point estimates.
- Regime transitions are detected with a lag (minimum duration filter); act on confirmed transitions, not single-day signals.
- The HMM is a generative model trained on historical data; future structural breaks (new regimes) may not be captured.
- This output is research / educational – not financial advice.

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